PHYSICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

26. The temperature of a body in air falls from 40°C to 24°C in 4 minutes. The temperature of the air is 16°C. The temperature of the body in the next 4 minutes will be :

(1) $\frac{14}{3}$°C (2) $\frac{28}{3}$°C

(3) $\frac{42}{3}$°C (4) $\frac{56}{3}$°C

Answer (4)

Sol. $\Delta T = T_2 - T_1 = 16°C$

And $T_0 = 16°C$

$$\frac{\Delta T}{t_1} = -k(32-16°) \quad ...(i)$$

$$\frac{(24-T_3)}{4} = -k\left(\frac{24+T_3}{2} - 16\right) \quad ...(ii)$$

$$\frac{16}{4} = -k(16)$$

$$\Rightarrow \frac{(24-T_3)}{4} = -k\left(12+\frac{T_3}{2}-16\right)$$

$$\Rightarrow \frac{16}{24-T_3} = \frac{16}{\frac{T_3}{T_2}-4}$$

$$\Rightarrow \frac{T_3}{2} - 4 = 24 - T_3$$

$$\Rightarrow \frac{3T_3}{2} = 28$$

$$\Rightarrow T_3 = \frac{56}{3}°C$$

27. A solid sphere is rolling without slipping on a horizontal plane. The ratio of the linear kinetic energy of the centre of mass of the sphere and rotational kinetic energy is :

(1) $\frac{5}{2}$ (2) $\frac{4}{3}$

(3) $\frac{3}{4}$ (4) $\frac{2}{5}$

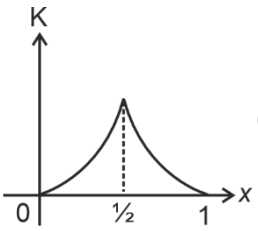
Answer (1)

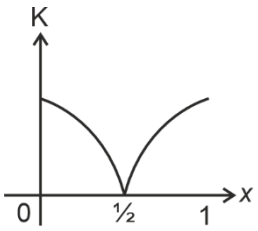
Sol. $KE_{(T)} = \frac{1}{2}mv^2$

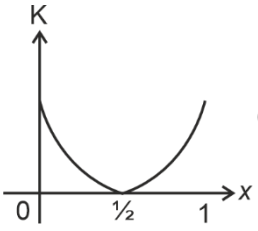
$$KE_{(R)} = \frac{1}{2}\cdot\frac{2}{5}mR^2\cdot\frac{v^2}{R^2} = \frac{1}{2}mv^2\left(\frac{2}{5}\right)$$

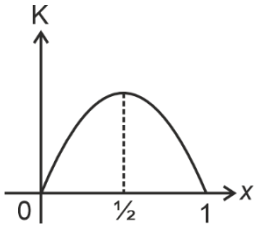
So, $\frac{KE_{(T)}}{KE_{(R)}} = \frac{5}{2}$

28. A particle oscillates along the *x*-axis according to the law, $x(t) = x_0 \sin^2\left(\frac{t}{2}\right)$ where x_0 = 1 m. The kinetic energy (K) of the particle as a function of *x* is correctly represented by the graph

(1)


(2)


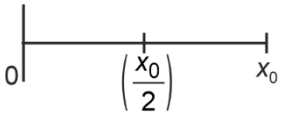
(3)


(4)


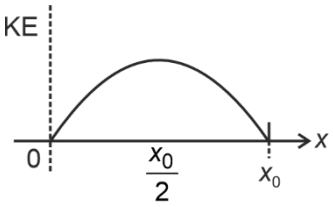
Answer (4)



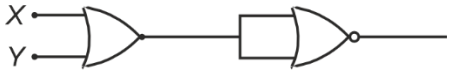
Sol. $x(t) = x_0 \sin^2\left(\frac{t}{2}\right) = \frac{x_0}{2}(1-\cos t)$



Clearly $\frac{x_0}{2}$ is mean position.



29. The output of the circuit is low (zero) for :



(A) $X = 0,\ Y = 0$ (B) $X = 0,\ Y = 1$

(C) $X = 1,\ Y = 0$ (D) $X = 1,\ Y = 1$

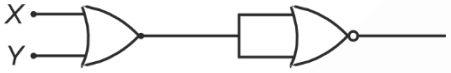
Choose the **correct** answer from the options given below :

(1) (A), (B) and (D) only (2) (A), (C) and (D) only

(3) (A), (B) and (C) only (4) (B), (C) and (D) only

Answer (4)

Sol.



Clearly $\overline{x+y}$ = output

So, when either of x or y is 1 then output will be zero.

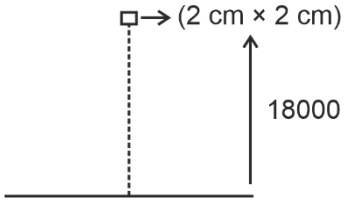
30. A photograph of a landscape is captured by a drone camera at a height of 18 km. The size of the camera film is 2 cm x 2 cm and the area of the landscape photographed is 400 km^2. The focal length of the lens in the drone camera is :

(1) 0.9 cm (2) 2.5 cm

(3) 1.8 cm (4) 2.8 cm

Answer (3)

Sol.



Linear size of image = 2 cm

Linear size of object = 20×10^5 cm

$$m = \frac{2}{20\times10^5} = 10^{-6}$$

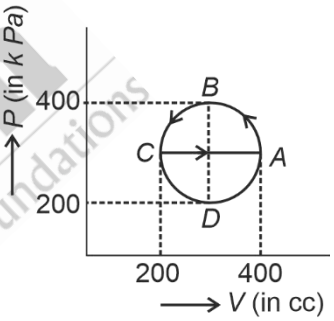
So, $v = \frac{u}{10^6}$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{10^6}{u} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{\left(1+10^6\right)}{u} = \frac{1}{f} \Rightarrow f \approx \left(\frac{u}{10^6}\right)$$

$$\Rightarrow f = \frac{18000\times100}{10^6}\ \text{cm} = 1.8\ \text{cm}$$

31. The magnitude of heat exchanged by a system for the given cyclic process *ABCA* (as shown in figure) is (in SI unit)



(1) 5 π

(2) 40 π

(3) 10 π

(4) Zero

Answer (1)

Sol. For cyclic process

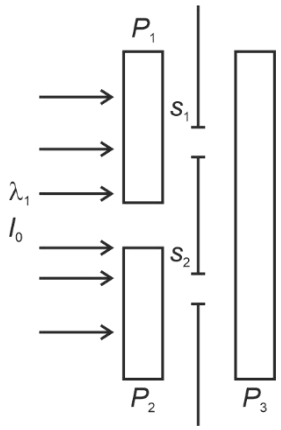
$\Delta u = \Delta Q - \Delta\omega = 0$

$$\Rightarrow |\Delta Q| = |\Delta\omega| = \frac{\pi}{2}\times100\times10^{-6}\times100\times10^3$$

$$\Rightarrow |\Delta Q| = 5\pi$$



32. In a Young's double slit experiment, three polarizers are kept as shown in the figure. The transmission axes of P_1 and P_2 are orthogonal to each other. The polarizer P_3 covers both the slits with its transmission axis at 45° to those of P_1 and P_2. An unpolarized light of wavelength λ and intensity I_0, is incident on P_1 and P_2. The intensity at a point after P_3 where the path difference between the light waves from s_1, and s_2 is $\frac{\lambda}{3}$, is



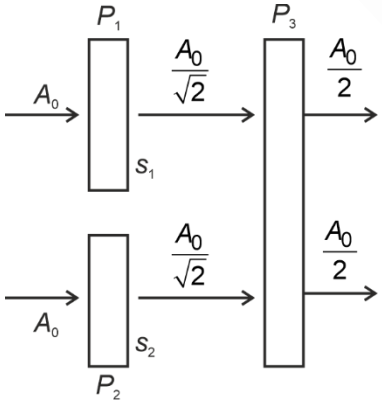
(1) $\frac{I_0}{3}$ (2) $\frac{I_0}{4}$

(3) $\frac{I_0}{2}$ (4) I_0

Answer (2)

Sol. Incident amplitude is A_0 then

$I_0 \propto A_0^2$



Now, path difference is $\frac{\lambda}{3}$ so phase difference is $\frac{2\pi}{3}$

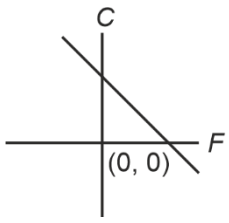
$A^2 = A_1^2 + A_2^2 + 2A_1A_2\cos\theta$

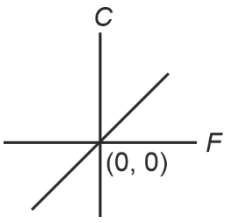
$= \frac{A_0^2}{4} + \frac{A_0^2}{4} + 2\frac{A_0^2}{4}\cdot\cos\left(\frac{2\pi}{3}\right)$

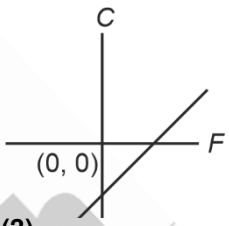
$\Rightarrow A^2 = \frac{A_0^2}{4}$

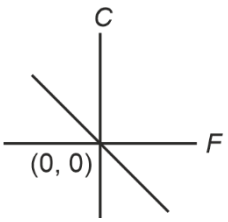
so, $I = \frac{I_0}{4}$

33. Which of the following figure represents the relation between Celsius and Fahrenheit temperatures?

(1) 

(2) 

(3) 

(4) 

Answer (3)

Sol. We know $F = 32 + \left(\frac{180}{100}\right)C$

$\Rightarrow \frac{180}{100}C = F - 32$

$\Rightarrow C = \left(\frac{100}{180}\right)F - \frac{160}{9}$

Clearly slope is +ve and intercept is –ve

34. Given below are two statements. One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : A electron in a certain region of uniform magnetic field is moving with constant velocity in a straight line path.

Reason (R) : The magnetic field in that region is along the direction of velocity of the electron.

In the light of the above statements, choose the correct answer from the options given below

(1) (A) is false but (R) is true

(2) Both (A) and (R) are true and (R) is the correct explanation of (A)

(3) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

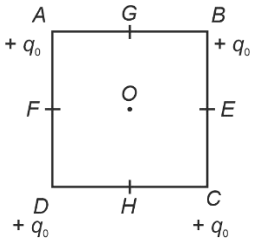
(4) (A) is true but (R) is false

Answer (2)

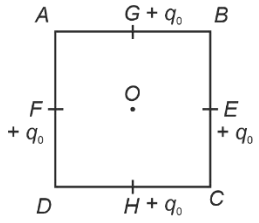


Sol. If the electron's velocity is along the direction of magnetic field then magnetic force on electron is zero and it will not accelerate.

35.



Configuration (1)



Configuration (2)

In the first configuration (1) as shown in the figure, four identical charges (q_0) are kept at the corners *A*, *B*, *C* and *D* of square of side length '*a*'. In the second configuration (2), the same charges are shifted to mid points G, E, H and F, of the square.

If $K = \frac{1}{4\pi\varepsilon_0}$, the difference between the potential energies of configuration (2) and (1) is given by

(1) $\frac{Kq_0^2}{a}\left(3\sqrt{2}-2\right)$ (2) $\frac{Kq_0^2}{a}\left(4\sqrt{2}-2\right)$

(3) $\frac{Kq_0^2}{a}\left(4-2\sqrt{2}\right)$ (4) $\frac{Kq_0^2}{a}\left(3-\sqrt{2}\right)$

Answer (1)

Sol. $u_① = \left(2\frac{Kq_0}{a}+\frac{Kq_0}{\sqrt{2}a}\right)q_0\times 2$

$u_② = \left(2\frac{Kq_0\sqrt{2}}{a}+\frac{Kq_0}{a}\right)q_0\times 2$

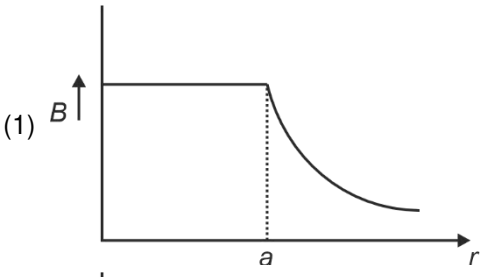
So, $\Delta u = u_2 - u_1 = 2q_0\frac{kq_0}{a}\left[2\sqrt{2}+1-2-\frac{1}{\sqrt{2}}\right]$

$\Rightarrow \Delta u = \frac{2q_0^2}{4\pi\varepsilon_0 a}\left[\frac{4-\sqrt{2}-1}{\sqrt{2}}\right] = \frac{2q_0^2}{4\pi\varepsilon_0 a}\frac{\left(3-\sqrt{2}\right)}{\sqrt{2}}$

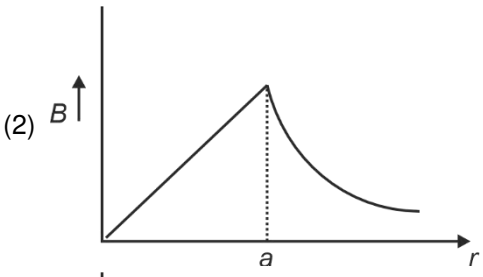
$\Rightarrow \Delta u = \frac{2kq_0^2}{a}\left[\frac{3-\sqrt{2}}{\sqrt{2}}\right] = \frac{kq_0^2}{a}\left(3\sqrt{2}-2\right)$

36. A long straight wire of a circular cross-section with radius '*a*' carries a steady current *I*. The current *I* is uniformly distributed across this cross-section. The plot of magnitude of magnetic field *B* with distance *r* from the centre of the wire is given by

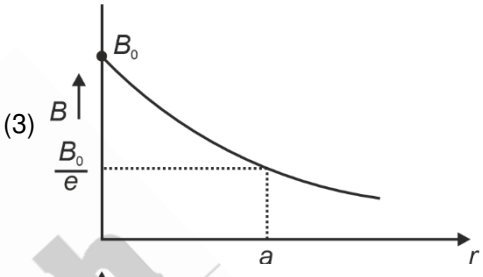
(1)



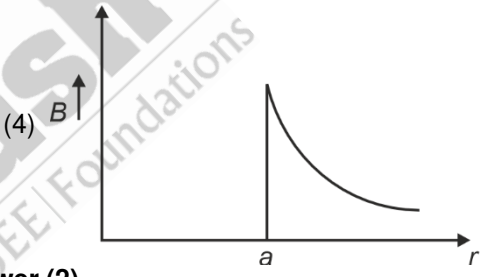
(2)



(3)



(4)



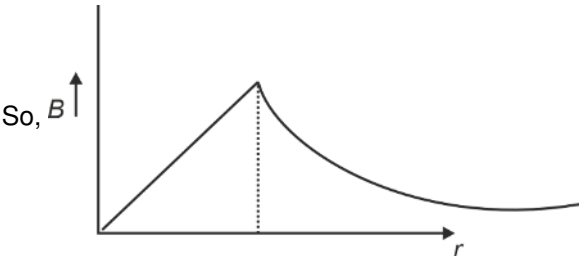
Answer (2)

Sol. We know inside the wire

$B = \frac{\mu_0 I}{2\pi R^2}\cdot r \qquad (O < r < R)$

And $B = \frac{\mu_0 I}{2\pi r}$ for $(R < r)$

So,



Option (2) is correct answer.



37 Young's double slit interference apparatus is immersed in a liquid of refractive index 1.44. It has slit separation of 1.5 mm. The slits are illuminated by a parallel beam of light whose wavelength in air is 690 nm. The fringe-width on a screen placed behind the plane of slits at a distance of 0.72 m, will be :

(1) 0.63 mm (2) 0.23 mm

(3) 0.33 mm (4) 0.46 mm

Answer (2)

Sol. $\Delta W = \frac{\lambda D}{d \cdot \mu} = \frac{690 \times 10^{-9} \times 0.72}{1.5 \times 10^{-3} \times 1.44}$

$\Rightarrow \Delta W = 2.3 \times 10^{-4}$ m = 0.23 mm

Option (2) is correct answer.

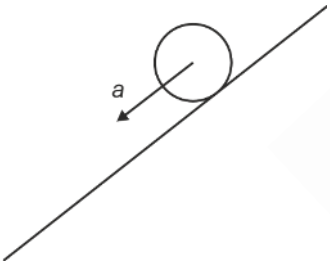
38. A solid sphere and a hollow sphere of the same mass and of same radius are rolled on an inclined plane. Let the time taken to reach the bottom by the solid sphere and the hollow sphere be t_1 and t_2, respectively, then

(1) $t_1 < t_2$ (2) $t_1 = 2t_2$

(3) $t_1 > t_2$ (4) $t_1 = t_2$

Answer (1)

Sol.



$mgR\sin\theta = I\frac{a}{R}$ $\Rightarrow$ $a = \frac{mgR^2\sin\theta}{I}$

For solid sphere $I_s = \frac{7}{5}mR^2$

So, $a_s = \frac{mgR^2\sin\theta}{7/5mR^2} = \frac{5g}{7}\sin\theta$

For $a_H = \frac{mgR^2\sin\theta}{\frac{5}{3}mR^2} = \frac{3}{5}g\sin\theta$

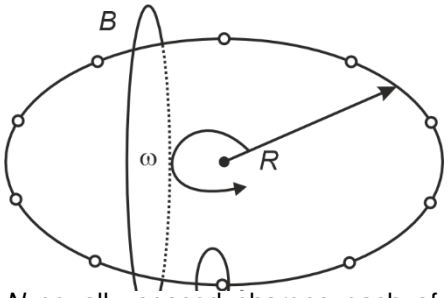
So, $\frac{1}{2}a_s t_1^2 = \frac{1}{2}a_H t_2^2$

$\Rightarrow \left(\frac{t_1}{t_2}\right)^2 = \frac{3\times7}{5\times5} = \frac{21}{25}$

Clearly $t_2 > t_1$

Option (1) is correct answer.

39.



N equally spaced charges each of value q, are placed on a circle of radius R. The circle rotates about its axis with an angular velocity ω as shown in the figure. A bigger Amperian loop B encloses the whole circle where as a smaller Amperian loop A encloses a small segment. The different enclosed currents, $I_A - I_B$, for the given Amperian loops is

(1) $\frac{N^2}{2\pi}q\omega$ (2) $\frac{N}{2\pi}q\omega$

(3) $\frac{2\pi}{N}q\omega$ (4) $\frac{N}{\pi}q\omega$

Answer (2)

Sol. $I = \frac{Nq\omega}{2\pi}$

Now for bigger Amperian loop (loop B)

$I_{B(\text{enclosed})} = 0$

And for smaller Amperian loop (loop A)

$|I_{A(\text{enclosed})}| = I = \frac{Nq\omega}{2\pi}$

So, $|I_A - I_B| = \frac{Nq\omega}{2\pi}$

Option (2) is correct answer.

40. A small uncharged conducting sphere is placed in contact with an identical sphere but having 4×10^{-8} C charge and then removed to a distance such that the force of repulsion between them is 9×10^{-3} N. The distance between them is (Take $\frac{1}{4\pi\varepsilon_0}$ as 9×10^9 in SI units)

(1) 3 cm (2) 2 cm

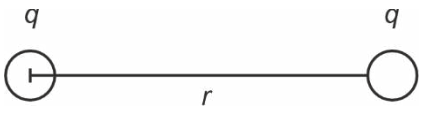
(3) 4 cm (4) 1 cm

Answer (2)



Sol. Charges will be shared equally

$$q = \frac{Q}{2}$$



$$F_{(rep)} = \frac{kq^2}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{r^2} = 9\times10^{-3}$$

$$\Rightarrow \quad \frac{9\times10^{9}\times4\times10^{-16}}{r^2} = 9\times10^{-3}$$

$$\Rightarrow \quad r^2 = \frac{4\times10^{-7}}{10^{-3}} = 4\times10^{-4}$$

$$\Rightarrow \quad r = 2\times10^{-2}\ \text{m} = 2\ \text{cm}$$

Option (2) is correct answer.

41. The energy E and momentum p of a moving body of mass m are related by some equation. Given that c represents the speed of light, identify the correct equation

(1) $E^2 = p^2c^2 + m^2c^4$

(2) $E^2 = p^2c^2 + m^2c^2$

(3) $E^2 = pc^2 + m^2c^4$

(4) $E^2 = pc^2 + m^2c^2$

Answer (1)

Sol. We need to check the dimensions only.

With momentum the dimension

$E^2 = p^2c^2$

And with mass $E^2 = m^2c^4$

So $E^2 = p^2c^2 + m^2c^4$ (dimensionally)

42. Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): In an insulated container, a gas is adiabatically shrunk to half of its initial volume. The temperature of the gas decreases.

Reason (R): Free expansion of an ideal gas is an irreversible and an adiabatic process.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

(2) (A) is true but (R) is false

(3) (A) is false but (R) is true

(4) Both (A) and (R) are true and (R) is the correct explanation of (A)

Answer (3)

Sol. If the container is insulated then temperature is expected to increase in adiabatic compression. So (A) is wrong.

And free expansion is irreversible and adiabatic process. So (R) is correct.

43. In photoelectric effect, the stopping potential (V_0) v/s frequency (ν) curve is plotted. (h is the Planck's constant and ϕ_0 is work function of metal)

(A) V_0 v/s ν is linear

(B) The slope of V_0 v/s ν curve = $\frac{\phi_0}{h}$

(C) h constant is related to the slope of V_0 v/s ν line

(D) The value of electric charge of electron is not required to determine h using the V_0 v/s ν curve.

(E) The work function can be estimated without knowing the value of h.

Choose the correct answer from the options given below :

(1) (A), (B) and (C) only

(2) (D) and (E) only

(3) (A), (C) and (E) only

(4) (C) and (E) only

Answer (3)



Sol. We know $h\nu - \phi_0 = eV_0$

$\Rightarrow V_0 = \left(\frac{h}{e}\right)\nu - \frac{\phi_0}{e}$

So slope is constant = $\frac{h}{e}$

and intercept is –ve $\left(\frac{\phi}{e}\right)$

$\Rightarrow$ Option (A) is correct, option (B) is wrong.

$\Rightarrow$ Option (C) is correct, option (D) is wrong.

$\Rightarrow$ Option (E) is correct.

(A), (C) & (E) are correct.

44. Arrange the following in the ascending order of wavelength (λ):

(A) Microwaves (λ_1)

(B) Ultraviolet rays (λ_2)

(C) Infrared rays (λ_3)

(D) X-rays (λ_4)

Choose the most appropriate answer from the options given below :

(1) $\lambda_3 < \lambda_4 < \lambda_2 < \lambda_1$ (2) $\lambda_4 < \lambda_3 < \lambda_1 < \lambda_2$

(3) $\lambda_4 < \lambda_3 < \lambda_2 < \lambda_1$ (4) $\lambda_4 < \lambda_2 < \lambda_3 < \lambda_1$

Answer (4)

Sol. X-ray (λ_4) is shortest wavelength

then ultraviolet is (λ_2) is longer than X-ray.

then infrared (λ_3) is longer than ultraviolet

then microwaves (λ_1) is longer than infrared

So $\lambda_4 < \lambda_2 < \lambda_3 < \lambda_1$

45. The position vector of a moving body at any instant of time is given as $\vec{r} = (5t^2\hat{i} - 5t\hat{j})$ m. The magnitude and direction of velocity at $t = 2$ s is,

(1) $5\sqrt{17}$ m/s, making an angle of $\tan^{-1}4$ with –ve Y axis

(2) $5\sqrt{17}$ m/s, making an angle of $\tan^{-1}4$ with +ve X axis

(3) $5\sqrt{15}$ m/s, making an angle of $\tan^{-1}4$ with +ve X axis

(4) $5\sqrt{15}$ m/s, making an angle of $\tan^{-1}4$ with –ve Y axis

Answer (1)

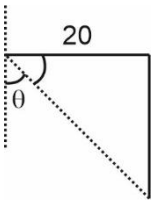
Sol. $\vec{r} = (5t^2\hat{i} - 5t\hat{i})$

So $\frac{d\vec{r}}{dt} = \vec{v} = (10t\hat{i} - 5\hat{j})$

At ($t = 2$ second)

$\vec{v}(t=2) = (20\hat{i} - 5\hat{j})$

$|\vec{v}| = 5\sqrt{1+16} = 5\sqrt{17}$ m/s

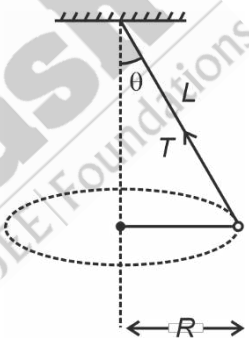


With –ve y-axis $\tan\theta = \frac{20}{5} = 4$

SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

46.



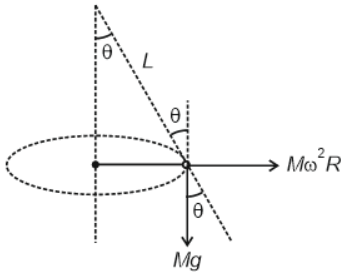
A string of length L is fixed at one end and carries a mass of M at the other end. The mass makes $\left(\frac{3}{\pi}\right)$ rotations per second about the vertical axis passing through end of the string as shown. The tension in the string is _____ ML.

Answer (36)

Sol. $\omega = \frac{3}{\pi} \times 2\pi = 6$ rad/s

$R = L\sin\theta$





and $T = M\sqrt{g^2 + \omega^4 R^2}$

Also, $T\sin\theta = M\omega^2 \cdot L\sin\theta$

$\Rightarrow T = M(36)\ L$

$\Rightarrow T = 36\ ML$

47. The ratio of the power of a light source S_1 to that the light source S_2 is 2. S_1 is emitting 2×10^{15} photons per second at 600 nm. If the wavelength of the source S_2 is 300 nm, then the number of photons per second emitted by S_2 is _____ $\times 10^{14}$.

Answer (5)

Sol. $p_1 = \dfrac{2\times10^{15}\times E_1}{1\ \text{sec}} = \dfrac{2\times10^{15}\times hc}{\lambda_1}$

$p_2 = \dfrac{N\times hc}{\lambda_2}$

$\dfrac{p_1}{p_2} = \dfrac{2\times10^{15}\times hc\times300}{600\times N\times hc} = 2$

$\Rightarrow N = \dfrac{10^{15}}{2} = 5\times10^{14}$ per second

48. The increase in pressure required to decrease the volume of a water sample by 0.2% is P $\times 10^5$ Nm^{-2}. Bulk modulus of water is 2.15 $\times 10^9$ Nm^{-2}. The value of P is _____.

Answer (43)

Sol. We know $\dfrac{\Delta P}{\Delta V/V} = -\beta$

$\Rightarrow \Delta P = -\beta\cdot\dfrac{\Delta V}{V}$

$\Rightarrow |\Delta P| = 2.15\times10^9 \times \dfrac{2}{1000} = 4.3\times10^6$ N/m^2

$\Rightarrow \Delta P = 43 \times 10^5$ N/m^2

49. Acceleration due to gravity on the surface of earth is 'g'. If the diameter of earth is reduced to one third of its original value and mass remains unchanged, then the acceleration due to gravity on the surface of the earth is _____ g.

Answer (9)

Sol. $g = \dfrac{GM}{R^2}$

Now, $g' = \dfrac{GM}{\left(\frac{R}{3}\right)^2} = \dfrac{9GM}{R^2} = 9g$

50. A tightly wound long solenoid carries a current of 1.5 A. An electron is executing uniform circular motion inside the solenoid with a time period of 75 ns. The number of turns per metre in the solenoid is _____.

[Take mass of electron $m_e = 9 \times 10^{-31}$ kg, charge of electron $|q_e| = 1.6 \times 10^{-19}$ C, $\mu_0 = 4\pi\times10^{-7}\dfrac{\text{N}}{\text{A}^2}$, 1 ns $= 10^{-9}$ s]

Answer (250)

Sol. We know,

$\beta = \mu_0\eta I$ and $T = \dfrac{2\pi m}{e\beta}$

$\Rightarrow \dfrac{2\pi m}{eT} = \mu_0\eta I$

$\Rightarrow \eta = \dfrac{2\pi m}{eT\mu_0 I}$

$= \dfrac{2\pi\times9\times10^{-31}}{1.6\times10^{-19}\times75\times10^{-9}\times4\pi\times10^{-7}\times1.5}$

$\Rightarrow \eta = \dfrac{9\times10^{-31}}{1.6\times10^{-19}\times75\times10^{-9}\times2\times10^{-7}\times1.5}$

$\Rightarrow \eta = \dfrac{9\times10^4}{360} = 250$ per meter



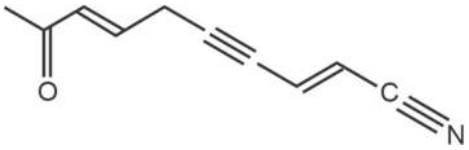
CHEMISTRY

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

51. In the given structure, number of sp and sp^2 hybridized carbon atoms present respectively are:

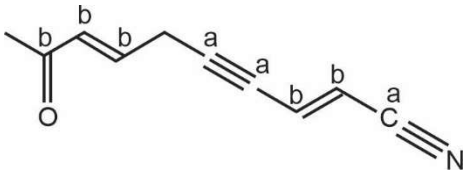


(1) 3 and 6 (2) 4 and 6

(3) 4 and 5 (4) 3 and 5

Answer (4)

Sol. The given structure is



Number of sp hybridized C-atoms(a) = 3

Number of sp^2 hybridized C-atoms(b) = 5

52. Identify correct statement/s:

(A) $-OCH_3$ and $-NHCOCH_3$ are activating group.

(B) –CN and –OH are meta directing group.

(C) –CN and $-SO_3H$ are meta directing group.

(D) Activating groups act as ortho- and para directing groups.

(E) Halides are activating groups.

Choose the correct answer from the options given below:

(1) (A), (C) and (D) only

(2) (A) and (C) only

(3) (A) only

(4) (A), (B) and (E) only

Answer (1)

Sol. $-OCH_3$, $NHCOCH_3$ and –OH are activating groups because the atom directly bonded to benzene ring activates the ring by +R effect using its lone pair of electrons. Activating groups are ortho- and para directing groups.

–CN and $-SO_3H$ are deactivating groups because the atom directly bonded to benzene is bonded to more electronegative atom through multiple bonds and they are meta directing groups due to –R effect. Halides are deactivating groups due to –I effect.

Correct statements are (A), (C) and (D) only.

53. The conditions and consequence that favours the $t_{2g}^3 e_g^1$ configuration in a metal complex are:

(1) strong field ligand, high spin complex

(2) weak field ligand, high spin complex

(3) weak field ligand, low spin complex

(4) strong field ligand, low spin complex

Answer (2)

Sol. The conditions and consequence that favour $t_{2g}^3 e_g^1$ configuration in a metal complex are

(i) weak field ligand, and

(ii) high spin complex

For weak field ligands, splitting energy (Δ_0) is lower than pairing energy (P). As a result distribution of electrons for $3d^4$ will be $t_{2g}^3 e_g^1$. It results in high spin complex due to maximum number of unpaired electrons.

